distribution over the θ_{ijk} parameters, with equal-valued parameters. If l is set to 1, this approach is called Laplace smoothing. See the chapter on Estimating Probabilites.

Maximum likelihood estimates for π_k are

$$\hat{\pi}_k = \hat{P}(Y = y_k) = \frac{\#D\{Y = y_k\}}{|D|} \tag{8}$$

where $|D|$ denotes the number of elements in the training set D.

Alternatively, we can obtain a smoothed estimate, or equivalently a MAP estimate based on a Dirichlet prior over the π_k parameters assuming equal priors on each π_k, by using the following expression

$$\hat{\pi}_k = \hat{P}(Y = y_k) = \frac{\#D\{Y = y_k\} + l}{|D| + lK} \tag{9}$$

where K is the number of distinct values Y can take on, and l again determines the strength of the prior assumptions relative to the observed data D.

2.4 Naive Bayes for Continuous Inputs

In the case of continuous inputs X_i, we can of course continue to use equations (2) and (3) as the basis for designing a Naive Bayes classifier. However, when the X_i are continuous we must choose some other way to represent the distributions $P(X_i|Y)$. One common approach is to assume that for each possible discrete value y_k of Y, the distribution of each continuous X_i is Gaussian, and is defined by a mean and standard deviation specific to X_i and y_k. In order to train such a Naive Bayes classifier we must therefore estimate the mean and standard deviation of each of these Gaussians:

$$\mu_{ik} = E[X_i|Y = y_k] \tag{10}$$

$$\sigma_{ik}^2 = E[(X_i - \mu_{ik})^2|Y = y_k] \tag{11}$$

for each attribute X_i and each possible value y_k of Y. Note there are $2nK$ of these parameters, all of which must be estimated independently.

Of course we must also estimate the priors on Y as well

$$\pi_k = P(Y = y_k) \tag{12}$$

The above model summarizes a Gaussian Naive Bayes classifier, which assumes that the data X is generated by a mixture of class-conditional (i.e., dependent on the value of the class variable Y) Gaussians. Furthermore, the Naive Bayes assumption introduces the additional constraint that the attribute values X_i are independent of one another within each of these mixture components. In particular problem settings where we have additional information, we might introduce additional assumptions to further restrict the number of parameters or the complexity of estimating them. For example, if we have reason to believe that noise in the observed X_i comes from a common source, then we might further assume that all of the σ_{ik} are identical, regardless of the attribute i or class k (see the homework exercise on this issue).

Again, we can use either maximum likelihood estimates (MLE) or maximum a posteriori (MAP) estimates for these parameters. The maximum likelihood estimator for μ_{ik} is

$$\hat{\mu}_{ik} = \frac{1}{\sum_j \delta(Y^j = y_k)} \sum_j X_i^j \delta(Y^j = y_k) \tag{13}$$

where the superscript j refers to the jth training example, and where $\delta(Y = y_k)$ is 1 if $Y = y_k$ and 0 otherwise. Note the role of δ here is to select only those training examples for which $Y = y_k$.

The maximum likelihood estimator for σ_{ik}^2 is

$$\hat{\sigma}_{ik}^2 = \frac{1}{\sum_j \delta(Y^j = y_k)} \sum_j (X_i^j - \hat{\mu}_{ik})^2 \delta(Y^j = y_k) \tag{14}$$

This maximum likelihood estimator is biased, so the minimum variance unbiased estimator (MVUE) is sometimes used instead. It is

$$\hat{\sigma}_{ik}^2 = \frac{1}{(\sum_j \delta(Y^j = y_k)) - 1} \sum_j (X_i^j - \hat{\mu}_{ik})^2 \delta(Y^j = y_k) \tag{15}$$

3 Logistic Regression

Logistic Regression is an approach to learning functions of the form $f : X \rightarrow Y$, or $P(Y|X)$ in the case where Y is discrete-valued, and $X = \langle X_1 \ldots X_n \rangle$ is any vector containing discrete or continuous variables. In this section we will primarily consider the case where Y is a boolean variable, in order to simplify notation. In the final subsection we extend our treatment to the case where Y takes on any finite number of discrete values.

Logistic Regression assumes a parametric form for the distribution $P(Y|X)$, then directly estimates its parameters from the training data. The parametric model assumed by Logistic Regression in the case where Y is boolean is:

$$P(Y = 1|X) = \frac{1}{1 + \exp(w_0 + \sum_{i=1}^n w_i X_i)} \tag{16}$$

and

$$P(Y = 0|X) = \frac{\exp(w_0 + \sum_{i=1}^n w_i X_i)}{1 + \exp(w_0 + \sum_{i=1}^n w_i X_i)} \tag{17}$$

Notice that equation (17) follows directly from equation (16), because the sum of these two probabilities must equal 1.

One highly convenient property of this form for $P(Y|X)$ is that it leads to a simple linear expression for classification. To classify any given X we generally

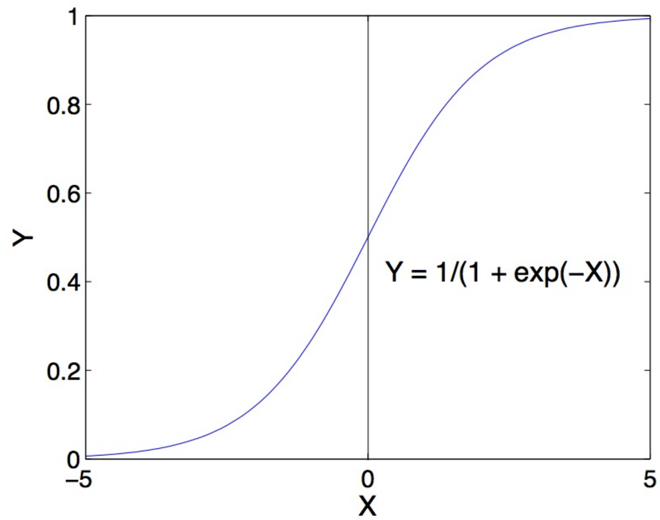

Figure 1: Form of the logistic function. In Logistic Regression, $P(Y|X)$ is assumed to follow this form.

want to assign the value y_k that maximizes $P(Y = y_k|X)$. Put another way, we assign the label $Y = 0$ if the following condition holds:

$$1 < \frac{P(Y=0|X)}{P(Y=1|X)}$$

substituting from equations (16) and (17), this becomes

$$1 < \exp(w_0 + \sum_{i=1}^{n} w_i X_i)$$

and taking the natural log of both sides we have a linear classification rule that assigns label $Y = 0$ if X satisfies

$$0 < w_0 + \sum_{i=1}^{n} w_i X_i \tag{18}$$

and assigns $Y = 1$ otherwise.

Interestingly, the parametric form of $P(Y|X)$ used by Logistic Regression is precisely the form implied by the assumptions of a Gaussian Naive Bayes classifier. Therefore, we can view Logistic Regression as a closely related alternative to GNB, though the two can produce different results in many cases.

3.1 Form of $P(Y|X)$ for Gaussian Naive Bayes Classifier

Here we derive the form of $P(Y|X)$ entailed by the assumptions of a Gaussian Naive Bayes (GNB) classifier, showing that it is precisely the form used by Logistic Regression and summarized in equations (16) and (17). In particular, consider a GNB based on the following modeling assumptions:

- Y is boolean, governed by a Bernoulli distribution, with parameter $\pi = P(Y=1)$
- $X = \langle X_1 \ldots X_n \rangle$, where each X_i is a continuous random variable
- For each X_i, $P(X_i|Y=y_k)$ is a Gaussian distribution of the form $N(\mu_{ik}, \sigma_i)$
- For all i and $j \neq i$, X_i and X_j are conditionally independent given Y

Note here we are assuming the standard deviations σ_i vary from attribute to attribute, but do not depend on Y.

We now derive the parametric form of $P(Y|X)$ that follows from this set of GNB assumptions. In general, Bayes rule allows us to write

$$P(Y=1|X) = \frac{P(Y=1)P(X|Y=1)}{P(Y=1)P(X|Y=1)+P(Y=0)P(X|Y=0)}$$

Dividing both the numerator and denominator by the numerator yields:

$$P(Y=1|X) = \frac{1}{1+\frac{P(Y=0)P(X|Y=0)}{P(Y=1)P(X|Y=1)}}$$

or equivalently

$$P(Y=1|X) = \frac{1}{1+\exp(\ln \frac{P(Y=0)P(X|Y=0)}{P(Y=1)P(X|Y=1)})}$$

Because of our conditional independence assumption we can write this

$$\begin{aligned} P(Y=1|X) &= \frac{1}{1+\exp(\ln \frac{P(Y=0)}{P(Y=1)} + \sum_i \ln \frac{P(X_i|Y=0)}{P(X_i|Y=1)})} \\ &= \frac{1}{1+\exp(\ln \frac{1-\pi}{\pi} + \sum_i \ln \frac{P(X_i|Y=0)}{P(X_i|Y=1)})} \end{aligned} \tag{19}$$

Note the final step expresses $P(Y=0)$ and $P(Y=1)$ in terms of the binomial parameter π.

Now consider just the summation in the denominator of equation (19). Given our assumption that $P(X_i|Y=y_k)$ is Gaussian, we can expand this term as follows:

$$\begin{aligned} \sum_i \ln \frac{P(X_i|Y=0)}{P(X_i|Y=1)} &= \sum_i \ln \frac{\frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(\frac{-(X_i-\mu_{i0})^2}{2\sigma_i^2}\right)}{\frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(\frac{-(X_i-\mu_{i1})^2}{2\sigma_i^2}\right)} \\ &= \sum_i \ln \exp\left(\frac{(X_i-\mu_{i1})^2-(X_i-\mu_{i0})^2}{2\sigma_i^2}\right) \\ &= \sum_i \left(\frac{(X_i-\mu_{i1})^2-(X_i-\mu_{i0})^2}{2\sigma_i^2}\right) \end{aligned}$$

$$
\begin{aligned}
&= \sum_i \left(\frac{(X_i^2 - 2X_i\mu_{i1} + \mu_{i1}^2) - (X_i^2 - 2X_i\mu_{i0} + \mu_{i0}^2)}{2\sigma_i^2} \right) \\
&= \sum_i \left(\frac{2X_i(\mu_{i0} - \mu_{i1}) + \mu_{i1}^2 - \mu_{i0}^2}{2\sigma_i^2} \right) \\
&= \sum_i \left(\frac{\mu_{i0} - \mu_{i1}}{\sigma_i^2} X_i + \frac{\mu_{i1}^2 - \mu_{i0}^2}{2\sigma_i^2} \right) \qquad (20)
\end{aligned}
$$

Note this expression is a linear weighted sum of the X_i's. Substituting expression (20) back into equation (19), we have

$$P(Y=1|X) = \frac{1}{1 + \exp(\ln\frac{1-\pi}{\pi} + \sum_i \left(\frac{\mu_{i0}-\mu_{i1}}{\sigma_i^2} X_i + \frac{\mu_{i1}^2-\mu_{i0}^2)}{2\sigma_i^2} \right))} \qquad (21)$$

Or equivalently,

$$P(Y=1|X) = \frac{1}{1 + \exp(w_0 + \sum_{i=1}^n w_i X_i)} \qquad (22)$$

where the weights $w_1 \ldots w_n$ are given by

$$w_i = \frac{\mu_{i0} - \mu_{i1}}{\sigma_i^2}$$

and where

$$w_0 = \ln\frac{1-\pi}{\pi} + \sum_i \frac{\mu_{i1}^2 - \mu_{i0}^2}{2\sigma_i^2}$$

Also we have

$$P(Y=0|X) = 1 - P(Y=1|X) = \frac{\exp(w_0 + \sum_{i=1}^n w_i X_i)}{1 + \exp(w_0 + \sum_{i=1}^n w_i X_i)} \qquad (23)$$

3.2 Estimating Parameters for Logistic Regression

The above subsection proves that $P(Y|X)$ can be expressed in the parametric form given by equations (16) and (17), under the Gaussian Naive Bayes assumptions detailed there. It also provides the value of the weights w_i in terms of the parameters estimated by the GNB classifier. Here we describe an alternative method for estimating these weights. We are interested in this alternative for two reasons. First, the form of $P(Y|X)$ assumed by Logistic Regression holds in many problem settings beyond the GNB problem detailed in the above section, and we wish to have a general method for estimating it in a more broad range of cases. Second, in many cases we may suspect the GNB assumptions are not perfectly satisfied. In this case we may wish to estimate the w_i parameters directly from the data, rather than going through the intermediate step of estimating the GNB parameters which forces us to adopt its more stringent modeling assumptions.

One reasonable approach to training Logistic Regression is to choose parameter values that maximize the conditional data likelihood. The conditional data likelihood is the probability of the observed Y values in the training data, conditioned on their corresponding X values. We choose parameters W that satisfy

$$W \leftarrow \arg\max_W \prod_l P(Y^l|X^l, W)$$

where $W = \langle w_0, w_1 \dots w_n \rangle$ is the vector of parameters to be estimated, Y^l denotes the observed value of Y in the lth training example, and X^l denotes the observed value of X in the lth training example. The expression to the right of the arg max is the conditional data likelihood. Here we include W in the conditional, to emphasize that the expression is a function of the W we are attempting to maximize.

Equivalently, we can work with the log of the conditional likelihood:

$$W \leftarrow \arg\max_W \sum_l \ln P(Y^l|X^l, W)$$

This conditional data log likelihood, which we will denote $l(W)$ can be written as

$$l(W) = \sum_l Y^l \ln P(Y^l = 1|X^l, W) + (1 - Y^l) \ln P(Y^l = 0|X^l, W)$$

Note here we are utilizing the fact that Y can take only values 0 or 1, so only one of the two terms in the expression will be non-zero for any given Y^l.

To keep our derivation consistent with common usage, we will in this section flip the assignment of the boolean variable Y so that we assign

$$P(Y = 0|X) = \frac{1}{1 + \exp(w_0 + \sum_{i=1}^n w_i X_i)} \tag{24}$$

and

$$P(Y = 1|X) = \frac{\exp(w_0 + \sum_{i=1}^n w_i X_i)}{1 + \exp(w_0 + \sum_{i=1}^n w_i X_i)} \tag{25}$$

In this case, we can reexpress the log of the conditional likelihood as:

$$\begin{aligned}
l(W) &= \sum_l Y^l \ln P(Y^l = 1|X^l, W) + (1 - Y^l) \ln P(Y^l = 0|X^l, W) \\
&= \sum_l Y^l \ln \frac{P(Y^l = 1|X^l, W)}{P(Y^l = 0|X^l, W)} + \ln P(Y^l = 0|X^l, W) \\
&= \sum_l Y^l (w_0 + \sum_i^n w_i X_i^l) - \ln(1 + \exp(w_0 + \sum_i^n w_i X_i^l))
\end{aligned}$$

where X_i^l denotes the value of X_i for the lth training example. Note the superscript l is not related to the log likelihood function $l(W)$.

Unfortunately, there is no closed form solution to maximizing $l(W)$ with respect to W. Therefore, one common approach is to use gradient ascent, in which we work with the gradient, which is the vector of partial derivatives. The ith component of the vector gradient has the form

$$\frac{\partial l(W)}{\partial w_i} = \sum_l X_i^l (Y^l - \hat{P}(Y^l = 1|X^l, W))$$

where $\hat{P}(Y^l|X^l,W)$ is the Logistic Regression prediction using equations (24) and (25) and the weights W. To accommodate weight w_0, we assume an imaginary $X_0 = 1$ for all l. This expression for the derivative has an intuitive interpretation: the term inside the parentheses is simply the prediction error; that is, the difference between the observed Y^l and its predicted probability! Note if $Y^l = 1$ then we wish for $\hat{P}(Y^l = 1|X^l,W)$ to be 1, whereas if $Y^l = 0$ then we prefer that $\hat{P}(Y^l = 1|X^l,W)$ be 0 (which makes $\hat{P}(Y^l = 0|X^l,W)$ equal to 1). This error term is multiplied by the value of X_i^l, which accounts for the magnitude of the $w_i X_i^l$ term in making this prediction.

Given this formula for the derivative of each w_i, we can use standard gradient ascent to optimize the weights W. Beginning with initial weights of zero, we repeatedly update the weights in the direction of the gradient, on each iteration changing every weight w_i according to

$$w_i \leftarrow w_i + \eta \sum_l X_i^l (Y^l - \hat{P}(Y^l = 1|X^l, W))$$

where η is a small constant (e.g., 0.01) which determines the step size. Because the conditional log likelihood $l(W)$ is a concave function in W, this gradient ascent procedure will converge to a global maximum. Gradient ascent is described in greater detail, for example, in Chapter 4 of Mitchell (1997). In many cases where computational efficiency is important it is common to use a variant of gradient ascent called conjugate gradient ascent, which often converges more quickly.

3.3 Regularization in Logistic Regression

Overfitting the training data is a problem that can arise in Logistic Regression, especially when data is very high dimensional and training data is sparse. One approach to reducing overfitting is *regularization*, in which we create a modified "penalized log likelihood function," which penalizes large values of W. One approach is to use the penalized log likelihood function

$$W \leftarrow \arg\max_W \sum_l \ln P(Y^l|X^l, W) - \frac{\lambda}{2}||W||^2$$

which adds a penalty proportional to the squared magnitude of W. Here λ is a constant that determines the strength of this penalty term.

Modifying our objective by adding in this penalty term gives us a new objective to maximize. It is easy to show that maximizing it corresponds to calculating the MAP estimate for W under the assumption that the prior distribution $P(W)$ is